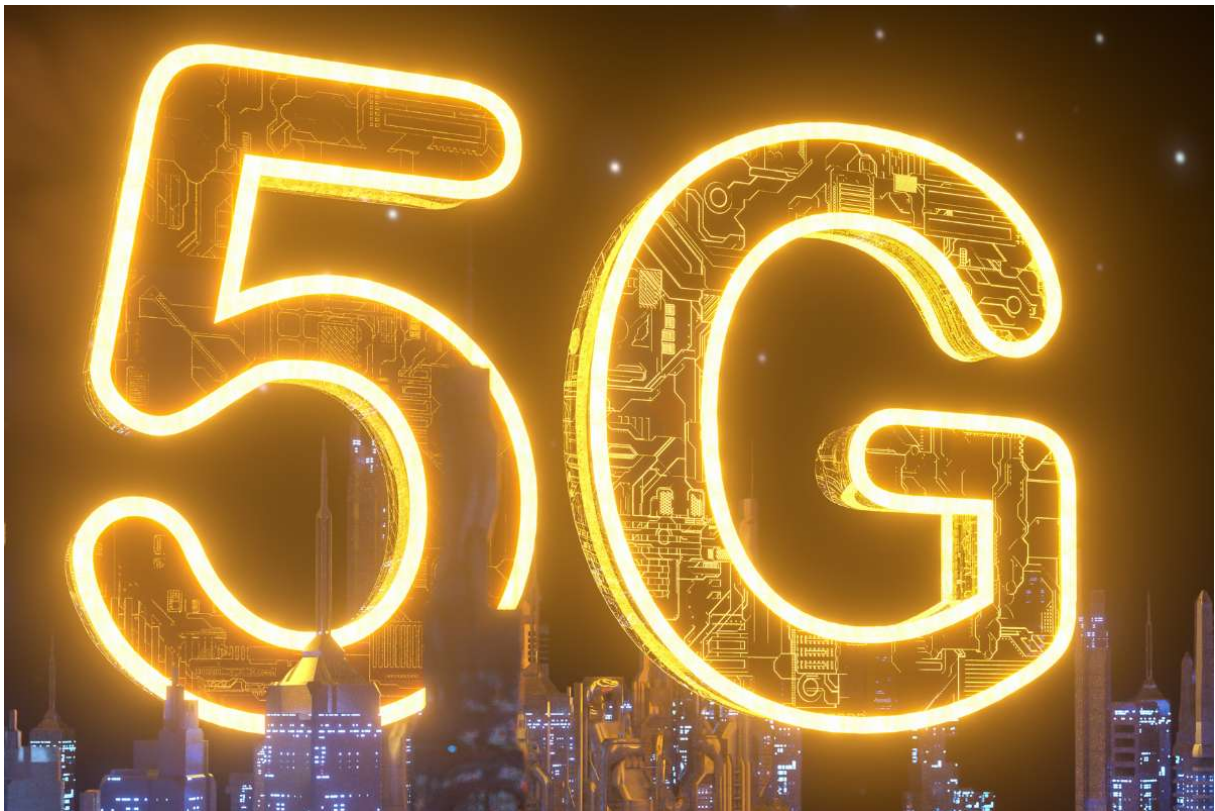


Project 9: 5G QoS Configuration and Service Differentiation Testing

**Encadrant : Serge FDIDA,
Thi-Mai-Trang NGUYEN,
Taha Mohsen ZAIDI**

**Etudiants : Kerrian AZIZA, Ihsane BOUBRIK,
Yasmine LEZOUL, Yanis YAHY**



All of the project code is available through GitHub with a SETUP_GUIDE.md explaining how to implement the tests and all of the project and a qos_configuration_guide.md for the qos profiles at the following address : <https://github.com/enashi/CELL>.

Table des matières

1	Introduction	3
2	5G QoS Architecture Overview	4
2.1	Quality of Service in 5G Networks	4
2.2	5QI Characteristics	4
2.3	QoS Handling in 5G Core	5
3	System Architecture	6
3.1	Overall Architecture	6
3.2	OpenAirInterface 5G Core Configuration	7
3.3	UERANSIM Configuration	7
4	QoS Profile Configuration	9
4.1	QoS Profiles Definition	9
4.2	Configured 5QI Profiles	9
4.3	QoS Flow Establishment Verification	10
5	Experiments	11
5.1	Experimental Setup	11
5.2	Traffic Generation Tools : iperf3	11
5.2.1	VoIP Traffic Generation	12
5.2.2	Video Streaming Traffic Generation	12
5.3	Performances Metrics	12
5.4	Application Testing Scenarios	13
5.4.1	Scenario 1 : VoIP Traffic	13
5.4.2	Scenario 2 : Video Streaming	14
5.5	Performance Analysis and Results	14
5.5.1	Phase 1 : Validation of the 5G Core Network Deployment	14
5.5.2	Phase 2 : Validation of QoS Configuration and 5QI Differentiation	15
5.5.3	Phase 3 : Scenario-Based Performance Evaluation Using JSON Results	16
6	Challenges and Limitations	19
6.1	Possible Future work	20
7	Conclusion	20

1 Introduction

The emergence of the fifth-generation mobile networks (5G) represents a significant evolution in wireless communication systems, promising enhanced capacity, ultra-low latency, and support for a massive number of connected devices. One of the critical enablers of these capabilities is the implementation of a flexible and efficient Quality of Service (QoS) framework, which allows network operators to differentiate traffic types according to their requirements and ensure that performance-sensitive applications, such as VoIP and real-time video streaming, meet strict latency and reliability constraints.

Unlike 4G networks, where QoS is primarily bearer-based with a limited set of standardized QCI values, 5G introduces QoS flows as the primary granularity of service differentiation. Each flow is identified by a 5G QoS Identifier (5QI), which defines key characteristics such as guaranteed or non-guaranteed bit rate (GBR/Non-GBR), priority level, packet delay budget (PDB), and packet error rate (PER). This framework allows for a more granular, flexible, and scalable approach to resource allocation, capable of supporting diverse use cases from enhanced mobile broadband (eMBB) to ultra-reliable low-latency communications (URLLC).

The management of QoS in 5G is tightly integrated into the 5G Core Network (5GC) architecture. User Equipment (UE) establishes PDU sessions via the gNB, which are managed by the Access and Mobility Management Function (AMF) and the Session Management Function (SMF). The SMF interacts with the Policy Control Function (PCF) and the User Plane Function (UPF) to enforce QoS policies, configure flows, and route traffic according to the assigned priorities. This end-to-end architecture ensures that the network can meet the specific service requirements for each application type.

This report focuses on the practical implementation and evaluation of 5G QoS in a controlled testbed environment. Using OpenAirInterface (OAI) 5G Core and the UERANSIM simulator, we configured multiple QoS flows for three representative traffic types : VoIP (high-priority GBR traffic), video streaming (medium-priority non-GBR traffic), and best-effort data traffic (low-priority non-GBR traffic). Each UE was configured with dedicated QoS profiles, specifying 5QI, priority levels, and ARP (Allocation and Retention Priority) parameters, ensuring differentiated handling of network resources and prioritization during congestion.

The experimental setup includes traffic generation using iperf3, validation of QoS flow establishment, and performance monitoring to assess throughput, latency, and packet loss for each service type. The results provide insights into the effectiveness of the QoS configuration and the ability of the 5G network to meet service-specific performance requirements. This approach also highlights the challenges and limitations inherent in implementing strict QoS differentiation in practical deployments.

The structure of this report is organized as follows : Section 2 provides an overview of the 5G QoS architecture, including the role of 5QI and QoS flow management. Section 3 details the system architecture and configuration of the OAI 5G Core and UERANSIM

simulator. Section 4 presents the QoS profile configuration, including the definition of 5QI profiles and verification of QoS flow establishment. Section 5 describes the experimental methodology, traffic generation, application testing scenarios, and performance metrics. Finally, Section 6 discusses the challenges, limitations, and potential future work, concluding with a summary of the key findings regarding QoS management in 5G networks.

2 5G QoS Architecture Overview

2.1 Quality of Service in 5G Networks

Quality of Service (QoS) in 5G networks has been designed to address the growing diversity of application requirements, ranging from real-time voice communication to high-throughput multimedia services and massive data transfers. Unlike previous generations, where traffic differentiation was relatively static and limited in granularity, 5G introduces a flexible and service-oriented QoS framework capable of supporting heterogeneous use cases with distinct performance constraints.

In 4G LTE networks, QoS is primarily managed through bearers associated with a limited set of standardized QoS Class Identifiers (QCI). Each bearer is configured with fixed parameters, which restricts the ability of the network to dynamically adapt to varying application demands. In contrast, the 5G system adopts a flow-based QoS model, where each service is handled through one or more QoS flows within a PDU session. This approach enables finer granularity in traffic differentiation and more efficient resource allocation across the network.

Each QoS flow is associated with specific performance requirements, such as latency, reliability, and throughput. These requirements are expressed through standardized QoS characteristics, allowing the 5G network to enforce differentiated treatment of packets at both the control plane and the user plane. As a result, applications with strict latency or reliability constraints, such as Voice over IP (VoIP), can be prioritized over less time-sensitive services, such as best-effort data traffic.

This flexible QoS framework is a key enabler for 5G service categories, including enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC). By decoupling service requirements from rigid bearer configurations, 5G allows network operators to adapt QoS policies to evolving application needs and traffic conditions.

2.2 5QI Characteristics

The 5G QoS Identifier (5QI) is a fundamental component of the 5G QoS framework, providing a standardized mechanism to describe the performance requirements of a QoS flow. Each 5QI value is associated with a predefined set of characteristics specified by the 3GPP, ensuring consistent interpretation and enforcement across different network implementations.

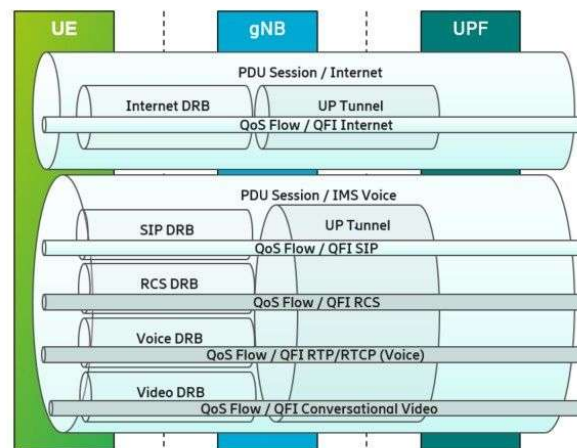


FIGURE 1 : Conceptual representation of a PDU session with multiple QoS flows

A 5QI defines several key parameters, including the resource type, priority level, packet delay budget (PDB), and packet error rate (PER). The resource type indicates whether the QoS flow is guaranteed bit rate (GBR), non-guaranteed bit rate (Non-GBR), or delay-critical GBR. GBR flows are allocated a minimum guaranteed throughput, making them suitable for real-time applications such as voice communication, while Non-GBR flows share network resources on a best-effort basis.

The priority level associated with a 5QI determines the relative importance of a QoS flow when competing for network resources. Lower numerical values correspond to higher priority, allowing critical services to be scheduled ahead of lower-priority traffic, particularly in congested scenarios. The packet delay budget specifies the maximum acceptable one-way latency for packet delivery, while the packet error rate defines the acceptable probability of packet loss.

In addition to standardized 5QI values, the 5G architecture allows the use of non-standardized or operator-specific 5QI values to support customized services. However, standardized 5QI values are generally preferred for interoperability and predictability. In this work, only standardized 5QI values are considered to ensure compliance with 3GPP specifications and to facilitate reproducible experimental evaluation.

By encapsulating multiple QoS characteristics into a single identifier, the 5QI simplifies QoS configuration while maintaining sufficient flexibility to support diverse service requirements. This abstraction plays a central role in the management of QoS flows throughout the 5G Core Network.

2.3 QoS Handling in 5G Core

The enforcement of QoS in 5G networks is tightly integrated into the 5G Core (5GC) architecture, which follows a service-based design. QoS management involves coordinated interactions between multiple network functions, ensuring that service requirements are

consistently applied from session establishment to packet forwarding.

When a User Equipment (UE) initiates a PDU session, the Access and Mobility Management Function (AMF) handles the signaling related to registration and mobility, while the Session Management Function (SMF) is responsible for establishing and managing the PDU session. The SMF retrieves subscription and policy information, including QoS parameters, from the Unified Data Management (UDM) and, when available, from the Policy Control Function (PCF).

Based on this information, the SMF determines the appropriate QoS flows and configures the User Plane Function (UPF) accordingly. The UPF is responsible for packet forwarding, traffic steering, and QoS enforcement at the user plane level. It applies scheduling and prioritization mechanisms to ensure that packets belonging to different QoS flows are treated according to their assigned 5QI and priority levels.

QoS flows are identified in the user plane by QoS Flow Identifiers (QFI), which allow the UPF to distinguish and process packets belonging to different services within the same PDU session. This design enables multiple applications with distinct QoS requirements to coexist within a single session, each benefiting from tailored network treatment.

Overall, the separation of control and user plane functions, combined with the flow-based QoS model, provides the 5G Core Network with a high degree of flexibility and scalability. This architecture allows dynamic adaptation of QoS policies in response to changing network conditions and service demands, forming the foundation for advanced service differentiation in 5G networks.

3 System Architecture

3.1 Overall Architecture

The system architecture implemented in this project is based on a virtualized 5G standalone (SA) network deployed in a controlled laboratory environment. The objective of this architecture is to evaluate the behavior of Quality of Service mechanisms within the 5G Core Network under different traffic conditions and service priorities. To achieve this, an open-source 5G Core implementation is combined with simulated radio access and user equipment components.

The architecture consists of three main elements : the 5G Core Network, simulated User Equipment (UE) and radio access components, and traffic generation and monitoring tools. All network functions are deployed as containerized services using Docker, enabling modularity, reproducibility, and simplified deployment. This approach allows the entire system to be executed on a single host machine while maintaining logical separation between network functions.

The 5G Core Network is implemented using OpenAirInterface (OAI), which provides an open-source realization of the 3GPP-defined 5GC architecture. The radio access network and user devices are emulated using UERANSIM, which simulates both the gNB and UE behavior and supports the establishment of PDU sessions with configurable QoS

parameters. Traffic is generated between UEs and external data networks using tools such as iperf3, enabling controlled performance evaluation.

This architecture enables end-to-end QoS evaluation, from PDU session establishment and QoS flow creation in the control plane to packet forwarding and prioritization in the user plane. It closely follows 3GPP specifications while remaining flexible enough to support experimentation and performance analysis.

3.2 OpenAirInterface 5G Core Configuration

The 5G Core Network in this project is implemented using the OpenAirInterface (OAI) 5G Core, deployed as a set of containerized network functions. Each core function is executed in a dedicated Docker container, allowing independent configuration and management. The main network functions involved in the architecture include the Access and Mobility Management Function (AMF), the Session Management Function (SMF), the User Plane Function (UPF), the Unified Data Management (UDM), the Authentication Server Function (AUSF), and the Network Repository Function (NRF).

The AMF is responsible for UE registration, authentication signaling, and mobility management, acting as the primary entry point for control-plane communication from the UE. The SMF manages PDU sessions and is the central entity for QoS flow configuration, as it determines the QoS parameters associated with each session based on subscription information and predefined policies. The UPF handles user-plane traffic, performing packet forwarding, traffic classification, and enforcement of QoS rules defined by the SMF.

The core network is configured to operate in standalone (SA) mode, without reliance on legacy LTE components. Network functions communicate using service-based interfaces, following the 5G Core service-based architecture. The UPF is connected to an external data network, allowing user traffic generated by UEs to reach external endpoints for performance testing.

Configuration files define network parameters such as IP addressing, interface bindings, and QoS-related policies. QoS enforcement is achieved by mapping QoS flows to specific traffic handling rules in the UPF, including prioritization and rate control mechanisms. This setup enables the differentiation of traffic types according to their assigned QoS profiles.

3.3 UERANSIM Configuration

UERANSIM is used in this project to emulate both the radio access network (gNB) and multiple User Equipment instances. It provides a lightweight and flexible simulation environment that supports essential 5G signaling procedures, including registration, PDU session establishment, and QoS flow management.

The simulated gNB acts as the access point between the UEs and the 5G Core Network. It establishes control-plane communication with the AMF and forwards user-plane traffic to the UPF via the N3 interface. Although the radio channel itself is not physically



FIGURE 2 : Overall architecture of the experimental 5G standalone testbed with a tunnel between 5G Core network in 192.168.70.0/24 and UERANSIM in 10.0.0.0/24

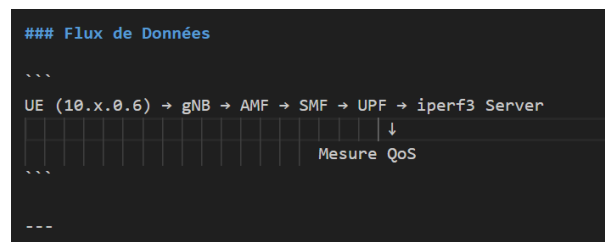


FIGURE 3 : Expected data flow in the system architecture.

modeled, UERANSIM accurately reproduces the protocol behavior required to validate core network functionality and QoS handling.

Each UE is configured with its own subscription profile, including identifiers, authentication credentials, and QoS-related parameters. UEs establish PDU sessions with the core network and are assigned specific QoS flows corresponding to different application types. This configuration allows multiple UEs to coexist simultaneously, each generating traffic with distinct QoS requirements.

Traffic generated by the UEs is forwarded through the gNB to the UPF and then to the external data network. By associating different traffic flows with different QoS profiles,

the system enables controlled evaluation of how the core network prioritizes and manages traffic under various conditions, including congestion scenarios.

The use of UERANSIM provides a practical compromise between realism and simplicity, allowing the focus to remain on core network behavior and QoS mechanisms rather than radio-level effects. This makes it particularly well suited for experimental studies of 5G QoS management in a laboratory environment.

4 QoS Profile Configuration

4.1 QoS Profiles Definition

The definition of QoS profiles is a critical step in enabling service differentiation within the 5G Core Network. In this project, QoS profiles are designed to reflect the requirements of representative application types with distinct performance constraints : real-time voice communication, multimedia streaming, and best-effort data traffic. Each profile is characterized by a specific set of QoS parameters, including the 5G QoS Identifier (5QI), priority level, resource type, and allocation and retention priority (ARP).

The selection of QoS parameters is guided by 3GPP specifications, ensuring that the configured profiles are aligned with standardized service characteristics. High-priority applications such as VoIP are assigned QoS profiles with strict latency and reliability constraints, whereas less time-sensitive applications are associated with non-guaranteed bit rate profiles. This approach allows the network to enforce differentiated treatment of traffic flows, particularly in congestion scenarios.

QoS profiles are associated with UEs at the time of PDU session establishment. Each UE is configured with subscription information that defines the default QoS parameters applied to its traffic. This static assignment enables controlled evaluation of QoS behavior and simplifies the interpretation of experimental results.

4.2 Configured 5QI Profiles

Based on the defined QoS profiles, standardized 5QI values are selected and configured in the 5G Core Network. The VoIP traffic is assigned a high-priority 5QI corresponding to GBR traffic with a low packet delay budget and strict packet error rate requirements. This configuration ensures minimal latency and reliable delivery, which are essential for real-time voice communication.

Video streaming traffic is associated with a medium-priority non-GBR 5QI. While video services require relatively high throughput, they are generally more tolerant to latency and packet loss than voice services. The selected 5QI reflects this balance by providing prioritized handling compared to best-effort traffic, without reserving guaranteed resources.

Best-effort data traffic is configured using a low-priority non-GBR 5QI. This profile does not guarantee throughput or latency and relies on available network resources. Such

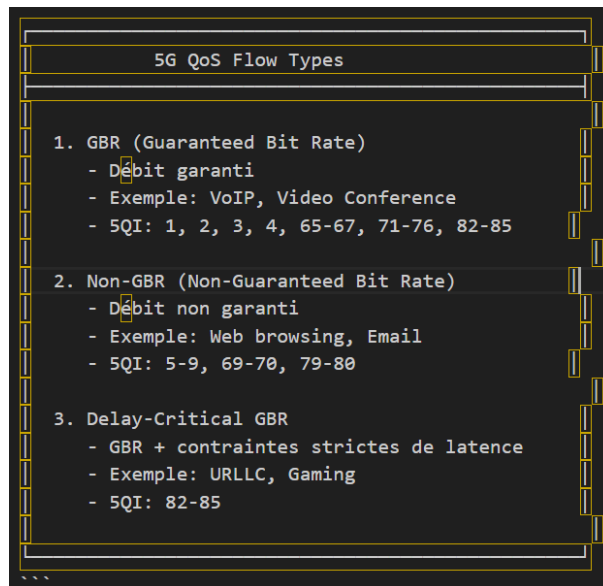


FIGURE 4 : Several QoS Profiles with 5QI

traffic is expected to experience performance degradation under congestion but serves as a useful baseline for evaluating QoS differentiation.

Each 5QI profile is mapped to a corresponding QoS Flow within the PDU session. These mappings are enforced by the Session Management Function, which configures the User Plane Function to apply appropriate scheduling and prioritization rules. By using standardized 5QI values, the configuration remains compliant with 3GPP recommendations and ensures predictable behavior across the network.

4.3 QoS Flow Establishment Verification

After configuring the QoS profiles and 5QI mappings, the correct establishment of QoS flows must be verified to ensure that the network applies the intended traffic differentiation. Verification is performed by observing signaling procedures during PDU session establishment and by inspecting runtime information provided by the 5G Core Network and simulation tools.

During PDU session establishment, the UE requests connectivity to the data network, triggering QoS flow creation by the Session Management Function. The SMF assigns one or more QoS Flow Identifiers (QFI) corresponding to the configured 5QI profiles. These QFIs are communicated to the UE and the gNB, enabling consistent identification of QoS flows across the network.

At the user plane level, the User Plane Function enforces the configured QoS rules by classifying packets based on their associated QFI. Logging information from the core network confirms that traffic flows are correctly mapped to their respective QoS profiles. This step is essential to ensure that observed performance differences in later experiments are the result of intentional QoS configuration rather than unintended behavior.

In addition to control-plane verification, practical validation is performed by generating traffic for each application type and observing flow behavior under normal and congested conditions. The successful differentiation of traffic performance confirms that QoS flows are correctly established and enforced throughout the network.

A complete explanation of the QoS Profile for our project is available in QOS_CONFIGURATION_GUIDE.md in the project github repo

5 Experiments

5.1 Experimental Setup

The experimental evaluation is conducted on a single host machine running a virtualized 5G standalone testbed. The host system executes the OpenAirInterface 5G Core Network using Docker containers and runs UERANSIM natively to simulate the gNB and multiple User Equipment instances. This setup enables controlled experimentation while preserving a realistic 5G Core behavior in accordance with 3GPP specifications.

Three UEs are instantiated simultaneously, each associated with a distinct application profile and QoS configuration. The first UE generates VoIP-like traffic and is configured with a high-priority guaranteed bit rate QoS profile. The second UE generates video streaming traffic and is assigned a medium-priority non-GBR profile. The third UE generates best-effort data traffic using a low-priority non-GBR profile. Each UE is connected to the same gNB and establishes a PDU session with the 5G Core Network.

Traffic generated by the UEs is forwarded through the gNB to the User Plane Function, which applies QoS enforcement and routes packets to an external data network. An iperf3 server, deployed as part of the test environment, is used as the traffic endpoint for throughput and latency measurements. This configuration allows direct observation of QoS differentiation effects under identical network conditions. All this architecture is as defined in Fig.2

5.2 Traffic Generation Tools : iperf3

Traffic generation is performed using iperf3, which provides flexible support for both TCP and UDP traffic patterns. This tool is well suited for evaluating network performance metrics such as throughput, packet loss, and jitter, and is widely used in experimental networking studies.

In this project, iperf3 is used to emulate different application behaviors. UDP traffic is employed to simulate real-time and streaming applications, allowing control over transmission rate and packet size. TCP traffic is used for best-effort data transfer, reflecting elastic applications such as file downloads and web browsing. The iperf3 server runs within the test environment, while clients are executed from the UE context to ensure that traffic traverses the full 5G Core Network.

The use of a single traffic generation tool across all scenarios ensures consistency in measurement methodology and simplifies comparison between different QoS profiles. Traffic generation parameters are carefully selected to stress the network and highlight differences in QoS handling, particularly under concurrent traffic conditions.

5.2.1 VoIP Traffic Generation

VoIP traffic is generated using a custom Python script, implemented in the file `test_voip.py` from the project scripts. This script simulates a G.711 VoIP codec by generating UDP packets at a constant rate of 50 packets per second, corresponding to a 20 ms packetization interval and a nominal bitrate of 64 kbps. This configuration reflects typical VoIP traffic characteristics.

The script performs multiple measurement steps, including baseline latency measurement using ICMP ping, real-time jitter measurement based on packet inter-arrival times, and packet loss estimation. Thresholds for acceptable performance are explicitly defined in the script, following ITU-T G.114 recommendations, with maximum acceptable values of 150 ms for latency, 30 ms for jitter, and 1% for packet loss. The script evaluates the measured performance against these thresholds and produces a pass/fail outcome.

5.2.2 Video Streaming Traffic Generation

Video streaming traffic is generated using the script `test_video.py` from the project scripts. This script primarily relies on `iperf3` in UDP mode to emulate continuous video streaming traffic at a target bitrate. Several predefined video quality profiles are supported, including standard definition (SD), high definition (HD), and higher bitrates, with the HD profile (5 Mbps) used in the conducted experiments.

As implemented in `test_video.py`, the `iperf3`-based approach is the recommended method, as it provides accurate throughput, packet loss, and jitter measurements. The script evaluates video streaming quality by comparing the achieved throughput to the target bitrate. A throughput ratio threshold of 95% is used to determine whether the test passes or fails. If `iperf3` is unavailable, the script includes a fallback socket-based traffic generation method, although this method is not used in the reported experiments.

5.3 Performances Metrics

The evaluation of the 5G QoS configuration is based on a set of performance metrics selected to reflect the requirements of the different application types considered in this study. These metrics are measured during traffic generation experiments and are used to assess the effectiveness of QoS differentiation under both isolated and mixed traffic conditions.

The following performance metrics are considered :

- **Throughput** : Throughput represents the effective data rate achieved during video streaming tests. It is measured using `iperf3`, as implemented in `test_video.py`, and

extracted from the iperf3 JSON output. Throughput is compared to the target bitrate to compute a throughput ratio, which serves as the primary performance indicator for video traffic.

- **Latency** : Latency is measured during VoIP tests using ICMP ping, as implemented in the function `measure_latency()` in `test_voip.py`. The average latency value is used to assess compliance with ITU-T G.114 recommendations.
- **Jitter** : Jitter is calculated during VoIP tests by measuring variations in packet inter-arrival times at the receiver. This measurement is implemented in the function `measure_jitter_live()` in `test_voip.py`, following principles defined in RFC 3550.
- **Packet Loss** : Packet loss is derived from ping statistics for VoIP tests and from iperf3 packet loss statistics for video streaming tests. Packet loss directly impacts perceived service quality and is therefore a critical metric for both traffic types.
- **Pass/Fail Outcome and QoS Score** : Each test script computes a pass/fail decision based on predefined thresholds and also derives a QoS score summarizing overall performance. These values are included in the result files and used for comparative analysis.

Together, these metrics provide a comprehensive view of network performance and enable objective comparison between different QoS profiles and traffic scenarios.

5.4 Application Testing Scenarios

To evaluate the behavior of the 5G QoS framework under realistic conditions, multiple testing scenarios are executed. Each scenario focuses on a specific traffic type or combination of traffic types and is designed to highlight the effect of QoS prioritization.

Scenario	Service Type	5QI	Traffic Pattern	Key Metrics
1	VoIP	1	Small packets, 50pps	Latency, jitter, loss
2	Video HD	9	Variable rate, 5Mbps avg	Throughput, buffering
3	Web Browsing	79	Bursty, HTTP requests	Page load time, latency
4	File Download	9	Bulk transfer, 10Mbps+	Throughput, completion time
5	Background	9	High volume	Throughput when deprioritized

FIGURE 5 : Table of all the mandatory tests to implement

5.4.1 Scenario 1 : VoIP Traffic

The VoIP testing scenario focuses exclusively on real-time voice traffic generated by a single UE. The test is executed using the `run_voip_test()` function in the file `test_voip.py`. During the test, VoIP packets are transmitted for a fixed duration, while latency, jitter, and packet loss are measured and evaluated.

The results of the VoIP test are stored in the file `results_voip.json`, which includes detailed statistics, threshold values, and an overall QoS score. These results confirm whether the configured high-priority QoS profile satisfies the strict performance requirements of real-time voice communication.

5.4.2 Scenario 2 : Video Streaming

The video streaming scenario evaluates the performance of continuous high-bitrate traffic generated by a UE configured with a medium-priority non-GBR QoS profile. The test is executed using the `run_video_test()` function implemented in `test_video.py`, with `iperf3` used as the traffic generation and measurement tool.

The test targets a predefined bitrate corresponding to an HD video profile. Achieved throughput, packet loss, and jitter are measured and stored in the file `results_video.json`. The script evaluates the results by computing a throughput ratio and comparing it against the predefined threshold. This scenario assesses whether the network can sustain video streaming quality under the configured QoS policy.

5.5 Performance Analysis and Results

The performance analysis is organized according to the different experimental phases defined in the project specification. This structured approach allows progressive validation of the system, starting from the correct deployment of the 5G Core Network, followed by verification of QoS configuration at the protocol level, and finally quantitative evaluation of application performance using traffic measurements.

5.5.1 Phase 1 : Validation of the 5G Core Network Deployment

The first phase consists in verifying that all required 5G Core Network functions are correctly deployed and operational. This validation step is essential to ensure that subsequent QoS experiments are conducted on a fully functional core network.

The deployment status is verified using the command `docker compose ps -a`, as shown in the project execution environment. The output confirms that all essential core network functions are running and in a healthy state. These functions include the Access and Mobility Management Function (AMF), Session Management Function (SMF), User Plane Function (UPF), Authentication Server Function (AUSF), Network Repository Function (NRF), Unified Data Management (UDM), and Unified Data Repository (UDR). In addition, supporting services such as the MySQL database and the `iperf3` server are also active.

The presence of the `iperf-server` container confirms that the traffic endpoint required for performance testing is available. The UPF container exposes user-plane interfaces required for data forwarding, while the AMF and SMF containers handle control-plane signaling and session management. All containers report a “healthy” status, indicating that the 5G Core Network is correctly initialized and ready for experimentation.

This phase validates the infrastructure layer of the project and confirms compliance with the deployment requirements defined in the project README.

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
iperf-server	networkstatic/iperf3	"iperf3 -s"	iperf-server	35 minutes ago	Up 35 minutes	0.0.0.0:5201->5201/tcp, [::]:5201->5201/tcp
mysql	mysql:8.0	"docker-entrypoint.s..."	mysql	35 minutes ago	Up 35 minutes (healthy)	3306/tcp, 33060/tcp
oai-smf	oaisoftwarealliance/oai-smf:develop	"/openair-smf/bin/oa..."	oai-smf	35 minutes ago	Up 35 minutes (healthy)	80/tcp, 5342-5344/tcp, 9090/tcp, 38412/sctp
oai-ausf	oaisoftwarealliance/oai-ausf:develop	"/openair-ausf/bin/oa..."	oai-ausf	35 minutes ago	Up 35 minutes (healthy)	80/tcp, 5342-5344/tcp
oai-nrf	oaisoftwarealliance/oai-nrf:develop	"/openair-nrf/bin/oa..."	oai-nrf	35 minutes ago	Up 35 minutes (healthy)	80/tcp, 5342-5344/tcp, 8080/tcp
oai-smf	oaisoftwarealliance/oai-smf:develop	"/openair-smf/bin/oa..."	oai-smf	35 minutes ago	Up 35 minutes (healthy)	80/tcp, 5342-5344/tcp, 9090/tcp, 8805/udp
oai-udm	oaisoftwarealliance/oai-udm:develop	"/openair-udm/bin/oa..."	oai-udm	35 minutes ago	Up 35 minutes (healthy)	80/tcp, 5342-5344/tcp
oai-udr	oaisoftwarealliance/oai-udr:develop	"/openair-udr/bin/oa..."	oai-udr	35 minutes ago	Up 35 minutes (healthy)	80/tcp
oai-upf	oaisoftwarealliance/oai-upf:develop	"sh /openair-upf/bin..."	oai-upf	35 minutes ago	Up 35 minutes (healthy)	2152/udp, 8805/udp, 5342-5344/tcp

FIGURE 6 : Containers corresponding to the different core network functions

5.5.2 Phase 2 : Validation of QoS Configuration and 5QI Differentiation

The second phase focuses on verifying that different User Equipment instances are effectively assigned distinct QoS profiles, as defined in the project configuration. This validation is performed at the protocol level using Wireshark captures of NGAP signaling during PDU session establishment.

Wireshark analysis of PDU Session Resource Setup Request messages reveals the presence of the QoSFlowSetupRequestList information element. Within this element, the nonDynamic5QI field explicitly indicates the 5QI value assigned to each QoS flow. The observed fiveQI values correspond directly to the QoS profiles configured for each UE in the project, confirming correct application of the configuration such as seen in Fig.??

The captured traces show that different UEs are associated with different 5QI values, reflecting their intended service types. For example, the UE configured for real-time traffic is assigned a high-priority 5QI, while UEs configured for video streaming or web traffic are assigned distinct non-GBR 5QI values. This confirms that QoS differentiation is enforced at the control-plane level by the Session Management Function.

In addition, the Wireshark traces confirm correct signaling paths within the 5G Core Network. The UE initiates PDU session establishment via the gNB, identified by the IP address 192.168.70.1. Control-plane communication is exchanged with the AMF and SMF, identified by the IP addresses 192.168.70.132 and 192.168.70.133, respectively. These observations are consistent with the expected 5G Core architecture and confirm correct interaction between network functions. This is seen in Fig.10

```

- Item 1: id-UL-NGU-IP-TNInformation
- ProtocolIE-Field
  id: id-UL-NGU-IP-TNInformation (139)
  criticality: reject (0)
  - value
    - UPTransportLayerInformation: gTPtunnel (0)
      - gTPtunnel
        - TransportLayerAddress: cba84686 [bit length 32, 1100 0000 1010 1000 0100 0110 1000 0110 decimal value 3232253574]
          TransportLayerAddress (IPv4): 192.168.70.134
          GTP-TEID: 00000000
- Item 2: id-PDUSessionType
- ProtocolIE-Field
  id: id-PDUSessionType (134)
  criticality: reject (0)
  - value
    - PDUSessionType: ipv4 (0)
- Item 3: id-QoSFlowSetupRequestList
- ProtocolIE-Field
  id: id-QoSFlowSetupRequestList (136)
  criticality: reject (0)
  - value
    - QoSFlowSetupRequestList: 1 item
      - Item 0
        - QoSFlowSetupRequestItem
          qosFlowIdentifier: 1
          qosFlowLevelQosParameters
            qosCharacteristics: nonDynamic5QI (0)
            nonDynamic5QI
              fiveQI: 1
          allocationAndRetentionPriority
            priorityLevelQoS: 1
            pre-emptionCapability: shall-not-trigger-pre-emption (0)
            pre-emptionVulnerability: not-pre-emptable (0)

```

FIGURE 7 : PDU Session Initial Request with 5qi corresponding to VoIP

Analysis Report

```

    • Item 0: id-AMF-UE-NGAP-ID
    • Item 1: id-AMF-UE-NGAP-ID
    • Item 2: id-POUSessionResourceSetupListSUReq
    - ProtocolIE-Field
      id: id-POUSessionResourceSetupListSUReq (74)
      criticality: reject (0)
      - value
        - POUSessionResourceSetupListSUReq: 1 item
        - Item 0
          - POUSessionResourceSetupItemSUReq
            qosFlowIdentifier: 1
            pduSessionNAS-PDU: 7e2b4b75e69d27e996801965324d810c211000901000631210100010060003296000a...
            s-NSSAI
            - POUSessionResourceSetupRequestTransfer: 00000400200000c320000020a00000000000a01f0ca846060000010006000210000...
            - POUSessionResourceSetupRequestTransfer
              - protocolIEs: 4 items
              - Item 0: id-POUSessionAggregateMaximumBitRate
                qosFlowIdentifier: 1
                id-UL-NGU-UP-TNInformation
              - Item 1: id-UL-NGU-UP-TNInformation
                id: id-POUSessionType
              - Item 2: id-POUSessionType
                id: id-qosFlowSetupRequestList
              - ProtocolIE-Field
                id: id-qosFlowSetupRequestList (136)
                criticality: reject (0)
                - value
                  - QosFlowSetupRequestList: 1 item
                  - Item 0
                    - QosFlowSetupRequestItem
                      qosFlowIdentifier: 1
                      qosFlowLevelQosParameters
                      - qosCharacteristics: nonDynamicQCI (0)
                      - nonDynamicQCI
                        fiveQI: 79
                      - allocationAndRetentionPriority

```

FIGURE 8 : PDU Session Initial Request with 5qi corresponding to Web traffic

```

    - protocolIEs: 4 items
    - Item 0: id-POUSessionAggregateMaximumBitRate
      - ProtocolIE-Field
        id: id-POUSessionAggregateMaximumBitRate (136)
        criticality: reject (0)
        - value
          - POUSessionAggregateMaximumBitRate
            qosFlowIdentifier: 1
            id-UL-NGU-UP-TNInformation
      - ProtocolIE-Field
        id: id-UL-NGU-UP-TNInformation (139)
        criticality: reject (0)
        - value
          - UPTransportLayerInformation: gTunnel (0)
    - Item 2: id-POUSessionType
      - ProtocolIE-Field
        id: id-POUSessionType (134)
        criticality: reject (0)
        - value
          - POUSessionType: ipv4 (0)
    - Item 3: id-qosFlowSetupRequestList
      - ProtocolIE-Field
        id: id-qosFlowSetupRequestList (136)
        criticality: reject (0)
        - value
          - QosFlowSetupRequestList: 1 item
          - Item 0
            - QosFlowSetupRequestItem
              qosFlowIdentifier: 1
              qosFlowLevelQosParameters
              - qosCharacteristics: nonDynamicQCI (0)
              - nonDynamicQCI
                fiveQI: 9
              - allocationAndRetentionPriority

```

FIGURE 9 : PDU Session Initial Request with 5qi corresponding to Video

This phase provides strong protocol-level evidence that each UE is associated with a distinct QoS flow and that QoS parameters are correctly propagated through the network.

No.	Time	Source	Destination	Protocol	Length	Info
224.2	0.79334051	192.168.70.132	192.168.70.1	NGAP/NL	136 SACK (Ack=1, Arwnd=104696)	DownLinkNASTransport, Security mode com.
225.2	0.81914033	192.168.70.1	192.168.70.132	NGAP/NL	186 SACK (Ack=1, Arwnd=104696)	UpLinkNASTransport
279.2	0.91315420	192.168.70.132	192.168.70.1	NGAP/NL	242 SACK (Ack=2, Arwnd=104696)	InitialContextSetupRequest
283.2	0.91749103	192.168.70.1	192.168.70.132	NGAP	98 SACK (Ack=2, Arwnd=104696)	InitialContextSetupResponse
285.2	0.94747119	192.168.70.1	192.168.70.132	NGAP/NL	234 UpLinkNASTransport, UpLinkNASTransport	
335.2	0.84835224	192.168.70.132	192.168.70.132	HTTP/2	868 DATA[1], JavaScript Object Notation (application/json), PDU session e.	
343.2	0.84835962	192.168.70.132	192.168.70.1	NGAP/NL	270 SACK (Ack=5, Arwnd=104696)	POUSessionResourceSetupRequest
344.2	0.84835911	192.168.70.1	192.168.70.132	NGAP	142 SACK (Ack=5, Arwnd=104696)	POUSessionResourceSetupResponse
2827.286	0.45177073	192.168.70.1	192.168.70.132	NGAP/NL	138 InitialUEMessage, Registration request	
2905.289	0.97863021	192.168.70.132	192.168.70.1	NGAP/NL	146 SACK (Ack=7, Arwnd=104696)	DownLinkNASTransport, Authentication req.
2906.289	0.97863037	192.168.70.1	192.168.70.132	NGAP/NL	146 SACK (Ack=4, Arwnd=104696)	UpLinkNASTransport, Authentication resp.
2980.289	0.980877069	192.168.70.132	192.168.70.1	NGAP/NL	126 SACK (Ack=9, Arwnd=104696)	DownLinkNASTransport, Security mode com.
2991.289	0.980877097	192.168.70.1	192.168.70.132	NGAP/NL	136 SACK (Ack=9, Arwnd=104696)	UpLinkNASTransport
3043.289	0.99013431	192.168.70.132	192.168.70.1	NGAP/NL	242 SACK (Ack=9, Arwnd=104696)	InitialContextSetupRequest
3047.289	0.99066232	192.168.70.1	192.168.70.132	NGAP	98 SACK (Ack=9, Arwnd=104696)	InitialContextSetupResponse
3052.289	0.99269072	192.168.70.1	192.168.70.132	NGAP/NL	234 UpLinkNASTransport, UpLinkNASTransport	
3362.289	0.11707007	192.168.70.132	192.168.70.132	HTTP/2	868 DATA[1], JavaScript Object Notation (application/json), PDU session e.	
3110.289	0.18054748	192.168.70.132	192.168.70.1	NGAP/NL	270 SACK (Ack=12, Arwnd=104696)	POUSessionResourceSetupRequest
3111.289	0.22669606	192.168.70.1	192.168.70.132	NGAP	142 SACK (Ack=7, Arwnd=104696)	POUSessionResourceSetupResponse
3030.244	0.52378083	192.168.70.1	192.168.70.132	NGAP/NL	138 InitialUEMessage, Registration request	
3030.244	0.56407847	192.168.70.132	192.168.70.1	NGAP/NL	146 SACK (Ack=10, Arwnd=104696)	DownLinkNASTransport, Authentication req.
3037.244	0.56578087	192.168.70.1	192.168.70.132	NGAP/NL	146 SACK (Ack=8, Arwnd=104696)	UpLinkNASTransport, Authentication resp.
3722.244	0.574666126	192.168.70.132	192.168.70.1	NGAP/NL	126 SACK (Ack=10, Arwnd=104696)	DownLinkNASTransport, Security mode com.
3722.244	0.575713565	192.168.70.1	192.168.70.132	NGAP/NL	136 SACK (Ack=9, Arwnd=104696)	UpLinkNASTransport
3786.244	0.58383393	192.168.70.132	192.168.70.1	NGAP	242 SACK (Ack=10, Arwnd=104696)	InitialContextSetupRequest
3781.244	0.582705566	192.168.70.1	192.168.70.132	NGAP	98 SACK (Ack=10, Arwnd=104696)	InitialContextSetupResponse
3782.244	0.702494932	192.168.70.1	192.168.70.132	NGAP/NL	234 UpLinkNASTransport, UpLinkNASTransport	
3833.244	0.70415208	192.168.70.132	192.168.70.132	HTTP/2	870 DATA[1], JavaScript Object Notation (application/json), PDU session e.	
3843.244	0.70605207	192.168.70.132	192.168.70.1	NGAP/NL	270 SACK (Ack=10, Arwnd=104696)	POUSessionResourceSetupRequest
3842.244	0.70610910	192.168.70.1	192.168.70.132	NGAP	142 SACK (Ack=11, Arwnd=104696)	POUSessionResourceSetupResponse

FIGURE 10 : Priority Level and Resource Allocation type depending on each Service Type

5.5.3 Phase 3 : Scenario-Based Performance Evaluation Using JSON Results

Visualization is used to support the interpretation of experimental results by providing a clear and intuitive representation of both QoS configuration and application-level performance. The figures presented in this subsection are generated automatically using the script report.figure.py, which processes the structured JSON result files produced during the experiments.

Figure 11 illustrates the QoS profile configuration applied to each service type. This figure highlights the relationship between service category, assigned 5QI, priority level,

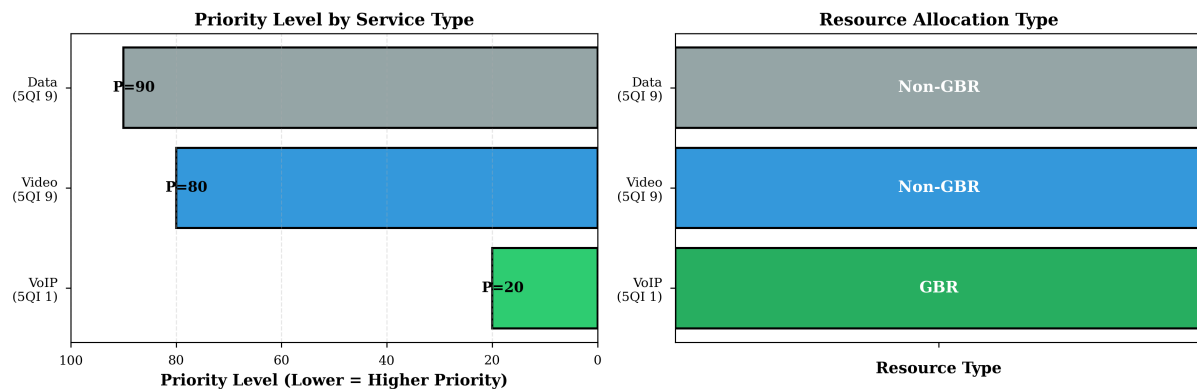


FIGURE 11 : Priority Level and Resource Allocation Type by Service Type

and resource allocation type. VoIP traffic is associated with 5QI 1 and a Guaranteed Bit Rate (GBR) profile, reflecting its real-time requirements and high scheduling priority. In contrast, video streaming and data traffic are assigned non-GBR profiles with higher priority values, consistent with their more elastic performance requirements.

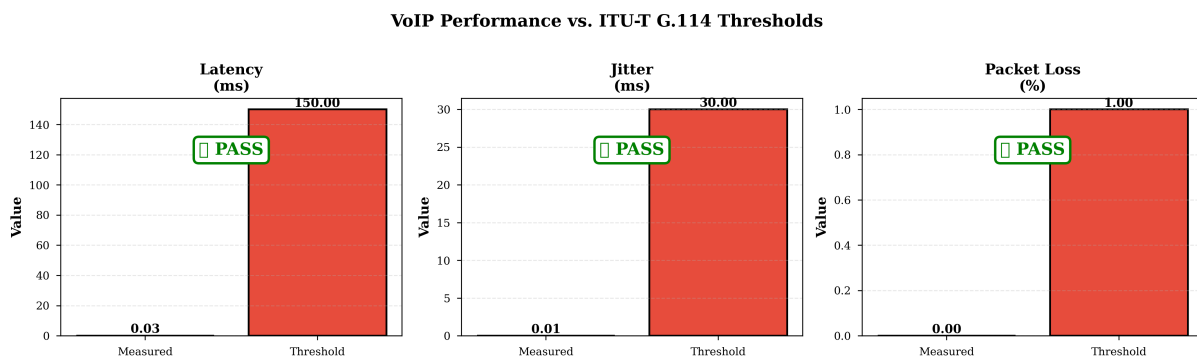


FIGURE 12 : Experimental results for the VoIP test compared to the official thresholds

VoIP performance is analyzed in Figure 12, which compares measured latency, jitter, and packet loss against ITU-T G.114 recommended thresholds. All measured values remain significantly below their respective limits, resulting in a successful classification of the VoIP scenario. This visualization provides standards-based validation that the QoS profile assigned to VoIP traffic effectively meets real-time communication requirements.

Figure 13 presents the performance results for video streaming traffic. The figure compares the target bitrate with the achieved throughput and the minimum acceptance threshold defined in the test configuration. The achieved throughput reaches 100% of the target bitrate, exceeding the required threshold of 95%, confirming that the video streaming QoS profile provides sufficient bandwidth under the tested conditions.

Finally, Figure 14 summarizes the overall experimental outcomes through a consolidated dashboard. This figure aggregates QoS scores, service priorities, and pass/fail outcomes for each scenario, providing a concise overview of the results. The dashboard

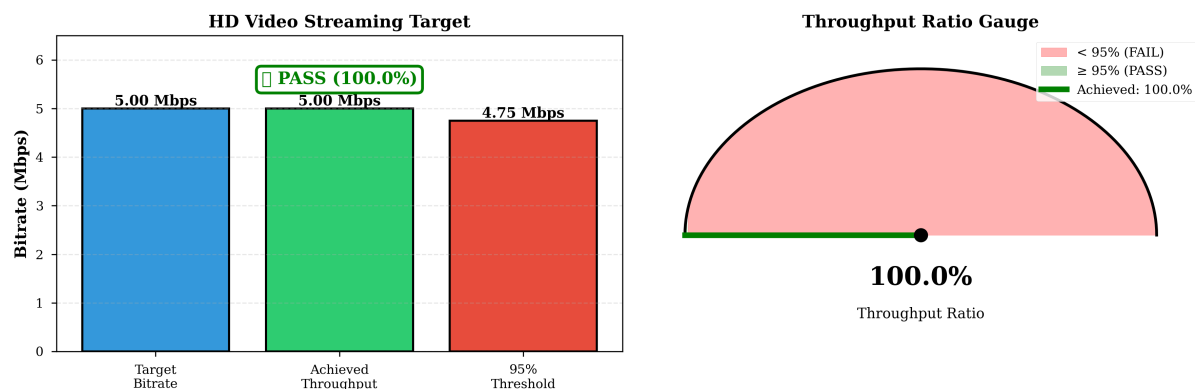
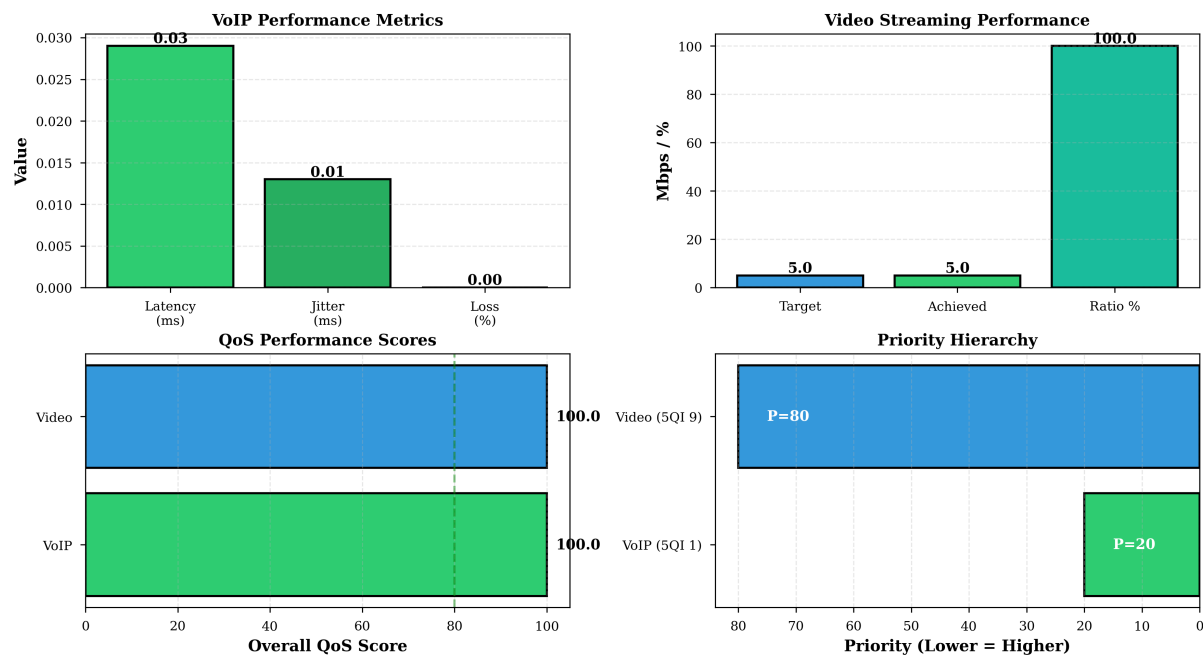


FIGURE 13 : Experimental results for the video test with Birate depending on the threshold and Throughput ratio Gauge



Test	5QI	Priority	Status	Score
VoIP Traffic	1 (GBR)	20 (High)	PASS	100.0/100
Video Streaming	9 (Non-GBR)	80 (Medium)	PASS	100.0/100

Summary: All Tests Passed Successfully

FIGURE 14 : Summary dashboard of the different tests with performances metrics

confirms that all tests are successfully completed and that the configured QoS profiles behave as expected in the current experimental environment.

6 Challenges and Limitations

Despite the successful implementation and validation of QoS differentiation in the 5G Core Network, several challenges and limitations must be acknowledged. These limitations are inherent to the experimental environment, the chosen tools, and the scope of the project, and they provide important context for interpreting the obtained results.

One of the primary limitations of this work is the use of a simulated radio access network. While UERANSIM accurately reproduces control-plane signaling and basic user-plane behavior, it does not model radio-level phenomena such as fading, interference, variable channel conditions, or mobility. As a result, the measured latency, jitter, and packet loss values are significantly lower and more stable than those typically observed in real-world 5G deployments. Consequently, the results primarily reflect core network behavior rather than end-to-end radio performance.

Another limitation concerns the scale of the experimental setup. The number of User Equipment instances and traffic flows is limited, which reduces the level of contention for network resources. Although mixed traffic scenarios are evaluated, the absence of heavy congestion and large numbers of concurrent UEs limits the ability to observe more complex QoS interactions, such as pre-emption effects or severe performance degradation for low-priority traffic.

The QoS configuration itself is static throughout the experiments. QoS profiles are pre-defined and assigned to UEs prior to PDU session establishment, and no dynamic adaptation of QoS parameters is performed during runtime. While this approach is sufficient to validate basic QoS differentiation, it does not reflect more advanced 5G capabilities such as dynamic policy control, network slicing, or real-time QoS adjustment based on network load or application behavior.

From a measurement perspective, the statistical analysis is constrained by the limited number of test repetitions. Although structured JSON result files and automated analysis scripts ensure reproducibility and consistency, the small sample size restricts the statistical significance of the results. Average values and pass/fail outcomes provide useful indicators but do not capture long-term variability or rare performance anomalies.

Finally, the evaluation focuses on technical performance metrics rather than user-perceived quality. While latency, jitter, throughput, and packet loss are appropriate indicators of network performance, they do not directly measure Quality of Experience (QoE) for end users. Metrics such as Mean Opinion Score (MOS) for VoIP or perceptual video quality metrics are not considered in this project.

Despite these limitations, the experimental setup remains well suited for validating the principles of QoS differentiation in a 5G Core Network. The combination of infrastructure validation, protocol-level inspection using Wireshark, and application-level performance measurements provides a coherent and rigorous evaluation framework. The identified limitations also highlight clear directions for future work and potential extensions of the project.

6.1 Possible Future work

Several extensions can be considered to further enhance the scope and realism of this project. A natural direction for future work would be to increase the scale of the experimental setup by introducing a larger number of User Equipment instances and concurrent traffic flows. This would allow more realistic congestion scenarios and enable deeper analysis of QoS behavior under heavy network load.

Another important extension would involve introducing dynamic QoS and policy control mechanisms. In the current implementation, QoS profiles are statically assigned to UEs at session establishment. Future work could integrate dynamic policy enforcement through the Policy Control Function (PCF), enabling real-time adaptation of QoS parameters based on network conditions, application behavior, or user subscription profiles.

The use of network slicing represents another promising area for further exploration. By combining QoS differentiation with network slicing, it would be possible to isolate resources for specific services or verticals, such as ultra-reliable communications or massive IoT applications. This would provide a more comprehensive evaluation of 5G service differentiation capabilities beyond individual QoS flows.

From an experimental perspective, replacing or complementing the simulated radio access network with real radio hardware would significantly improve realism. Incorporating physical gNBs and UEs would allow the evaluation of radio-level impairments, mobility, and handover procedures, providing a more complete end-to-end performance assessment.

Finally, future work could extend performance evaluation beyond network-centric metrics by incorporating Quality of Experience (QoE) indicators. Metrics such as Mean Opinion Score (MOS) for voice services or perceptual video quality measures would provide a closer link between network performance and user-perceived service quality.

These extensions would build upon the foundations established in this project and contribute to a more comprehensive understanding of QoS management in 5G networks.

7 Conclusion

This project investigated the implementation and evaluation of Quality of Service mechanisms in a 5G standalone network, with a particular focus on service differentiation at the core network level. Using an open-source 5G Core Network based on OpenAirInterface and a simulated radio access network provided by UERANSIM, a complete experimental testbed was deployed to study how different traffic types are handled according to their QoS requirements.

The work began with the successful deployment and validation of the 5G Core Network infrastructure. All essential network functions, including the AMF, SMF, and UPF, were correctly instantiated and verified, ensuring a stable foundation for experimentation. This infrastructure-level validation was a crucial prerequisite for reliable QoS evaluation.

QoS differentiation was then implemented by assigning distinct QoS profiles to different User Equipment instances. Each UE was configured with a specific 5QI correspon-

ding to its intended application type. Protocol-level inspection using Wireshark confirmed that these configurations were correctly applied during PDU session establishment, with distinct QoS flows and 5QI values visible in NGAP signaling messages. This validation step provided concrete evidence that QoS differentiation was not only configured but also actively enforced within the 5G Core Network.

Application-level performance evaluation further demonstrated the effectiveness of the configured QoS profiles. VoIP traffic achieved very low latency and jitter, meeting real-time communication requirements, while video streaming traffic sustained its target throughput with minimal packet loss. The use of structured JSON result files and automated analysis scripts enabled consistent and reproducible performance assessment, linking protocol-level configuration to observable service behavior.

While the experimental environment has inherent limitations, particularly due to the use of a simulated radio access network and a limited number of traffic flows, the results clearly demonstrate the core principles of QoS management in 5G networks. The project successfully illustrates how standardized 5QI-based QoS mechanisms can be implemented, validated, and evaluated in practice using open-source tools.

Overall, this work provides a coherent and practical exploration of QoS in 5G Core Networks, bridging theoretical concepts defined by 3GPP with hands-on experimentation. It highlights both the potential and the challenges of QoS enforcement in next-generation mobile networks and lays a solid foundation for future work involving more dynamic policies, larger-scale scenarios, and advanced performance evaluation techniques.